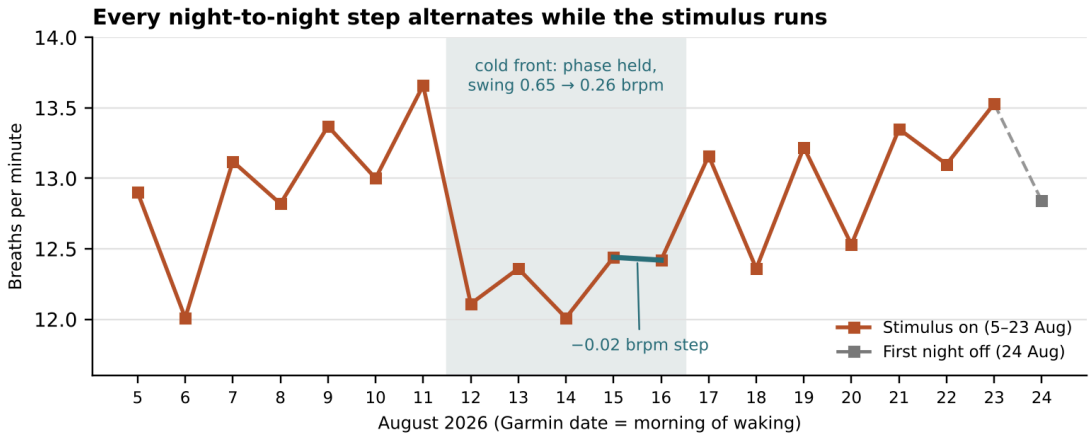
Nightingale’s thermally modulated tonal drops: a night-to-night zigzag in sleep breathing rate and a multi-month transition fingerprint

Single-subject observations, Highveld, South Africa · August–October 2026 · Garmin nightly averages

While the stimulus runs, nightly sleep breathing rate alternates high–low from one night to the next. August is the statistical test: 19 nights, every step alternating. The later periods are qualitative: the alternation breaks when the stimulus or the daily routine is interrupted, and resumes with the stimulus, from the first night after a 400 km move to a setting with different ambient temperatures.

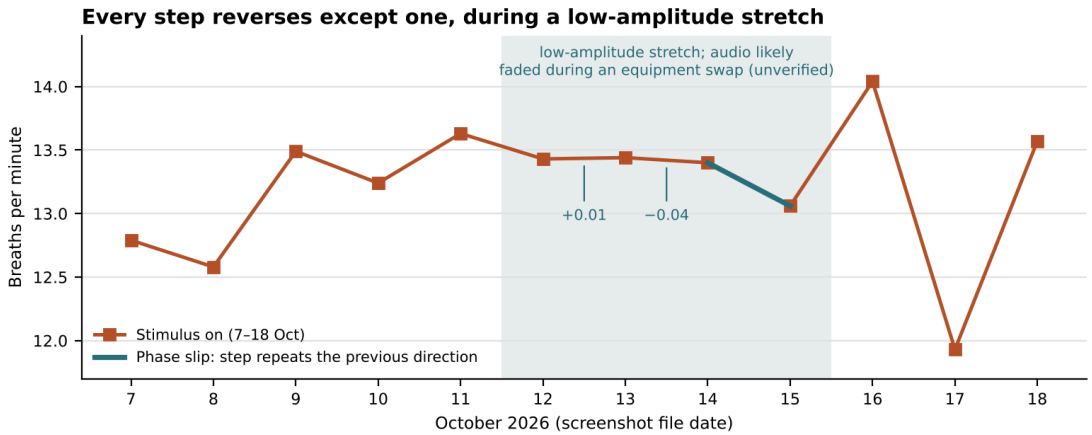
> **What Nightingale is.** Nightingale grew out of Nightbheat, the author’s water-cooling sleep system (a 2019 finalist in the Innovation Hub’s GAP Medical competition). A small pump circulates cool water through a mat under the thighs, close to the large leg blood vessels. The returning water falls into the water tank in the regime between a stream and single drops, giving soft tonal “drops” in the 1–5 kHz range through the night. Because the water passes under the body before it falls, the sleeper’s own heat, posture, breathing and pulse shape its flow and temperature; temperature changes the water’s viscosity, and viscosity changes the pitch of the drops. The sleeper is gently cooled while hearing a faint acoustic trace of their own physiology. It is mains-powered with UPS backup, non-invasive, and needs no action from the user. The working hypothesis is that this combined thermal and tonal input helps re-tune night-time autonomic regulation; the mechanism has not been tested.

1 · August 2026, city: 19 stimulated nights, every step alternating



Stimulus on 5–23 August; 24 August is the first night off (the step into it still reversed; the first step with both nights off did not). Shaded: a cold front, during which the swing shrank from 0.65 to 0.26 brpm and the phase held. The 15→16 August step is −0.02 brpm, below meaningful resolution. $p = 4.8 \times 10^{-4}$ for either phase, assuming independent nights.

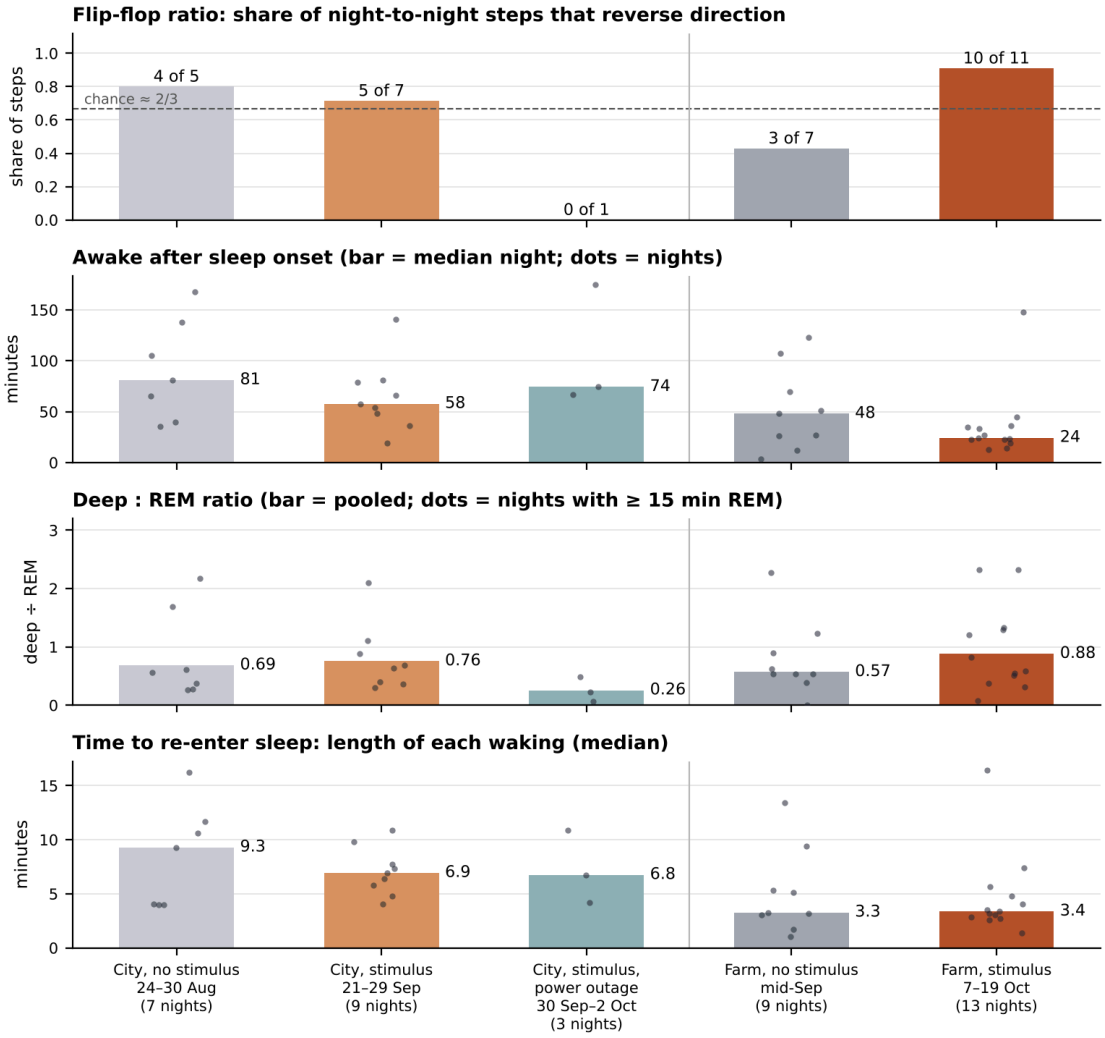
2 · October 2026, farm: 12 nights, one phase slip



From 7 October every step reverses except 14→15, which follows two steps below meaningful resolution (+0.01, −0.04 brpm). **Possible cause (not verified):** the audio signal faded during an equipment swap; an audio recorder will run alongside from now on. The run starts after a **confirmed UPS outage on the night of 4 October** that stopped the stimulus and the water cooling; that night had 5.6% deep sleep, 10.2% REM and 97 min awake, though 2 October, with no known outage, looked similar. Screenshot readings (August: Garmin app averages).

Farm, family household, irregular timing, hot nights. Qualitative: the 7 October start was chosen after seeing the data, so no probability is claimed. References on the same farm without the stimulus: 9 nights in mid-September 2026 (3 of 7 steps reversing, longest run 3 nights) and 17 nights in October 2025 (no long alternating runs).

3 · Five conditions, August–October 2026 (qualitative)



Within each place, nights with the stimulus had less time awake after sleep onset than nights without: city 58 vs 81 min (same window room, the week after the August run), farm 24 vs 48 min. Each waking was shortest on the farm either way; in the city it fell from 9.3 min without the stimulus to 6.9 min with it. The flip-flop ratio is highest on the farm with the stimulus (10 of 11) and lowest there without it (3 of 7); in the city, the week after the August run still reversed on 4 of 5 steps, so the alternation may carry over for some days after stimulation ends. Deep : REM stays between 0.57 and 0.88 except during the outage (0.26). Screenshot readings, 3 to 13 nights per condition; no tests claimed (city awake time on vs off: p ≈ 0.35).

4 · Stops and restarts, August–October 2026

Nights	Place	Conditions	Night-to-night breathing
5–23 Aug	City	Stimulus on	Every step alternates, 19 nights (section 1)
24–30 Aug	City	Stimulus off	Step into 24 Aug reverses; the first step with both nights off does not, nor does 26→27; 27–30 Aug reverse again without the stimulus. Awake time 81 min vs 36 min in the second half of the run
Mid-Sep, 9 nights	Farm	No stimulus	No zigzag: 3 of 7 steps reverse, longest run 3 nights
21–24 Sep	City, after the move	Stimulus on, settling in	Steps alternate by sign, but two are below resolution (−0.02, −0.06)
25–30 Sep	City	Stimulus on	Every step alternates, 6 nights; high nights on odd dates, as in August
From 29 Sep	City	Power outage, stimulus on UPS; no evening shower, less cycling, 11–12 h in bed	Step into the outage reverses, the next two do not, as at the end of August
4 Oct	—	Confirmed UPS failure: stimulus and cooling off	Run of section 2 begins after it
7–19 Oct	Farm	Stimulus on	12 nights alternate, one slip (section 2)
20 Oct	Relocation	Stimulus off	No reversal: breathing falls on both steps around the off night
21–24 Oct	**City, 400 km from the farm**	**Stimulus on again; different ambient temperatures**	**Alternates from the first night: 21 high, 22 low, 23 high, 24 low (steps −0.36, +0.85, −0.60 brpm)**

Screenshot readings except August. Each segment is too short to test alone (four alternating nights occur by chance 42% of the time); the point is that every documented interruption coincides with a break and every restart with a resumption, including from the first night after the 400 km move. The outage suggests the alternation also depends on the daily routine (evening shower, exercise).

5 · The transition fingerprint: sleep continuity before and after settling

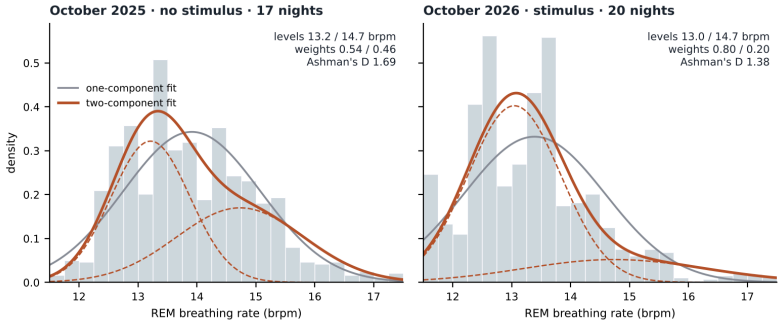
Median per night	Aug 2026 nights 1–8	Aug 2026 nights 9–17	Oct 2026 (farm)	Nights 1–8 vs Oct (p)	Nights 9–17 vs Oct (p)
Awake after first falling asleep	85 min	36 min	24 min	**0.006**	0.06
Sleep efficiency	88%	94%	95%	**0.010**	0.20
Wake bouts per hour	1.39	1.11	1.00	**0.033**	0.27
Stage switches per hour	4.01	3.51	3.20	**0.016**	0.19
Length of each waking	6.8 min	4.6 min	3.3 min	**0.013**	0.31
Breathing rate	12.49 brpm	12.33 brpm	13.27 brpm	0.033	0.04
Sleep	503 min	535 min	464 min	0.033	0.003

Mann-Whitney tests, 8 and 9 nights against 20. Continuity (first five rows) converged with October by late August; the grey rows depend on season and routine. Awake time fell across August 2026 ($\rho = -0.55$, $p = 0.02$), not in August 2025 without the stimulus. Late August was in the city and October on the farm, so location may contribute.

6 · REM breathing bimodality: history, and a possible stabilising role for Nightingale

Nightingale did not cause the reduction in REM bimodality: that came with the first intervention in December 2025 and was already present on the farm in 2026 without Nightingale. It may help sustain the reduction. The data so far neither show nor rule this out.

Period	Conditions	REM breathing structure
Oct 2022, Mar 2025, Apr 2025	Baseline, 46 nights	86–94% of REM segments bimodal, levels about 13.3 / 14.3 brpm, concentrated 02:00–05:00 (scored by eye; Ashman's D 2.36–2.40)
4–20 Oct 2025	Farm, no intervention (watch lent to a family member from 21 Oct 2025 to Jan 2026)	Two balanced levels, 13.2 / 14.7 brpm, weights 0.54 / 0.46 (left panel below)
Mid-Dec 2025 – Apr 2026	First intervention: mechanical stochastic resonance via the sleep surface, then electrolytes	Bimodal fraction 35% (Mar), then 0% (Apr), by eye
Jul 2026	Washout without the vibrating pillow	Two levels partly return, about 13.0 / 14.3 brpm (automated D ≈ 1.47)
Aug 2026	City, Nightingale starts	Two balanced levels, weights 0.49 / 0.51
24–30 Aug 2026	City, Nightingale off	Two levels, weights 0.61 / 0.39
Mid-Sep 2026	Farm, no Nightingale	One dominant level, weights 0.69 / 0.31
21–29 Sep 2026	City, Nightingale	Two balanced levels, weights 0.50 / 0.50
7–19 Oct 2026	Farm, Nightingale	One dominant level, weights 0.68 / 0.32; whole month 0.80 / 0.20 (right panel below)

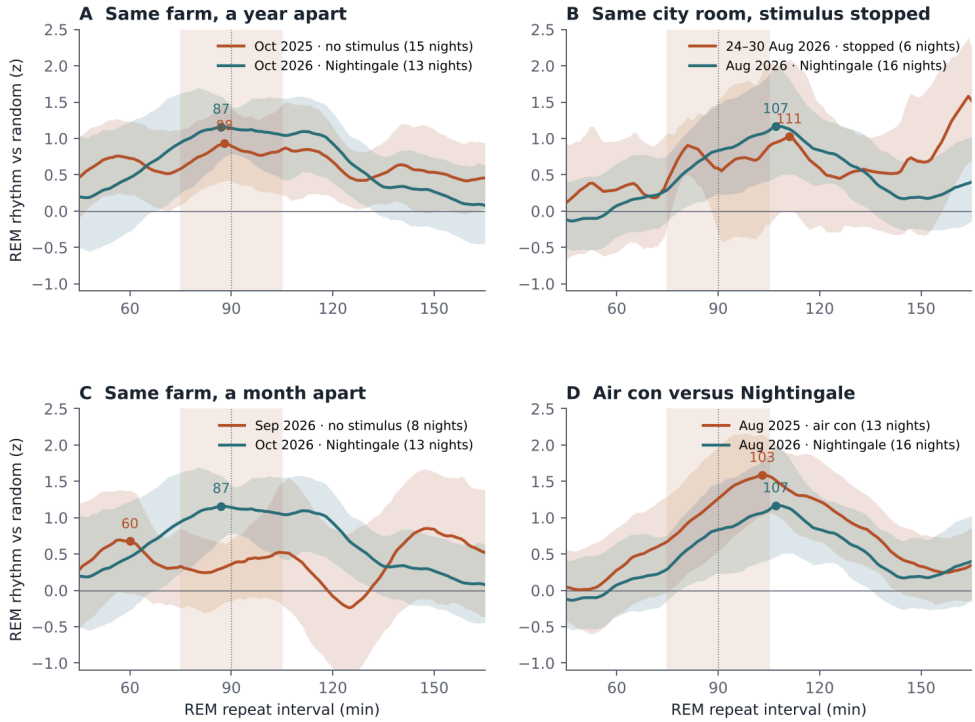


Same farm and foam mattress, no electrolytes in either year. REM samples ≥ 11.5 brpm; REM episodes in the last hour before waking excluded; automated screenshot extraction.

Reading it: in 2026 the two-level pattern follows the place more than Nightingale. On the farm it is reduced with and without Nightingale; in the city the second level is present with and without it (0.61 / 0.39 off, about 0.50 / 0.50 on). So Nightingale does not by itself hold the reduction against the city room and its larger day–night temperature swings. Whether it helps sustain the reduction on the farm cannot yet be told, since the farm looked the same with and without it. The test is a farm block with Nightingale off for one to two weeks, with sleep posture logged, to see whether the second level returns.

7 · REM ultradian rhythm: four paired comparisons

How regularly REM sleep returns through the night, in pairs that hold one thing constant. With Nightingale on the farm, REM recurs most clearly near 90 minutes. The same farm without it a month earlier shows almost no rhythm.



Each curve is the autocorrelation of Garmin's REM labels (1-minute resolution), expressed against the same night's REM episodes placed at random (z, mean across nights; shaded band = 95% bootstrap interval). Dots mark the strongest repeat between 60 and 130 minutes. Beige band 75–105 min.

Pair	Strongest repeat	Rhythm at 90 min (z, 95% interval)
A · Oct 2025 farm, no stimulus vs Oct 2026 farm, Nightingale	88 vs 87 min	0.90 (0.41–1.40) vs 1.14 (0.79–1.45)
B · 24–30 Aug 2026 city, stopped vs Aug 2026 city, Nightingale	111 vs 107 min	0.59 (−0.30–1.53, 6 nights) vs 0.84 (0.24–1.47)
C · Sep 2026 farm, no stimulus vs Oct 2026 farm, Nightingale	none clear vs 87 min	0.36 (−0.15–0.69) vs 1.14 (0.79–1.45)
D · Aug 2025 city, air con vs Aug 2026 city, Nightingale	103 vs 107 min	1.26 (0.54–1.97) vs 0.84 (0.24–1.47)

Reading it: pair C is the clearest: same farm, a month apart, and the intervals do not overlap. In A and B the Nightingale curve is the stronger, but the intervals overlap. In D, air con gives a rhythm at least as strong, so Nightingale looks comparable to air con on this measure, not better. Farm nights sit near 90 minutes and city nights run longer cycles, with or without the stimulus. Limits: Garmin infers REM from wrist signals; REM start times alone do not lock to a 90-minute clock in any period; season and room differ within pairs A and D.

How to read it

- **Counting rule:** every step counts by its sign, however small; exactly equal nights break a run. August uses Garmin app nightly averages; later periods are read from sleep-timeline screenshots (they agree: 24 Aug, 12.84 vs 12.82 brpm). With any minimum step size the longest August run is 11 nights.
- **What it is:** one person, one device, consumer wrist data. The alternation largely tracks the stimulus, with breaks at documented interruptions and some carry-over after it stops, which makes it testable; the carrier is unknown.
- **What it is not yet:** the 8/9 split was chosen after seeing the data, spring warming is not excluded, and nothing here has been reproduced by anyone else. The next test is pre-registered on/off blocks within one season, with an audio recorder running alongside.

Data, scoring rule and working documents: github.com/hartmutbohmer/Nightbheat